ALEXANDER C. JENKINS

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ABOUT ME

I'm a theoretical physicist, working at the interface between *cosmology*, *high-energy physics*, and *quantum matter*. My research looks at new ways of probing the fundamental laws of Nature, whether that's using *gravitational waves* as cosmic messengers, or using *quantum technologies* to simulate the early Universe.

EMPLOYMENT

Gavin Boyle Fellow in Cosmology — *University of Cambridge* *2024–present*
EPSRC Stephen Hawking Fellow *2025–present*
Research fellow hosted in the Kavli Institute for Cosmology Cambridge (KICC) and the Department of Applied Mathematics and Theoretical Physics (DAMTP) | Fellow of Selwyn College

Postdoctoral Research Fellow — *University College London* *2021–2024*
Led an international, interdisciplinary project to study vacuum decay with quantum analogue experiments and lattice simulations, as part of the QSimFP Consortium | Member of the Cosmoparticle Initiative

EDUCATION

PhD in Theoretical Physics — *King's College London* *2017–2021*
Funded by competitive faculty scholarship | Theoretical Particle Physics and Cosmology Group
'*Cosmology and fundamental physics in the era of GW astronomy*', supervised by Mairi Sakellariadou

MSci in Astrophysics (Part III) — *University of Cambridge* *2016–2017*
1st class | Ranked 5th in cohort | Elected a Bateman Scholar of Trinity Hall for 'excellent exam results'

Leonid Grishchuk Internship Programme — *Cardiff University* *Summer 2016*
Funded summer research internship in the Gravitational Physics Group

MA in Natural Sciences (Astrophysics) — *University of Cambridge* *2013–2016*
1st class | Elected a Scholar of Trinity Hall

GRANTS AND FUNDING SECURED (POST-PHD)

- EPSRC Stephen Hawking Fellowship (PI, £506k fEC) *2025*
  3-year research council fellowship supporting 'visionary scientists working in theoretical physics'
  Proposal ranked 2nd out of 26 by EPSRC prioritisation panel
- Gavin Boyle Fellowship, KICC and Selwyn College, Cambridge | 5-year institutional fellowship *2024*
- UKRI Quantum Technologies for Fundamental Physics Grant extension (Co-I, £69k) *2023*
  Wrote successful proposal for 6-month funded extension to my QSimFP research

AFFILIATIONS

Scientific collaborations
∗ Quantum Simulators for Fundamental Physics (*2021–present*) ∗ LISA Consortium (*2018–present*)
∗ Einstein Telescope (*2021–present*) ∗ LIGO Scientific Collaboration (*2016–2021*)

Professional bodies
∗ Institute of Physics (MInstP; *2018–present*) ∗ Royal Astronomical Society (FRAS; *2020–present*)
∗ Junior Member, European Astronomical Society (*2020–present*)

AWARDS AND ACHIEVEMENTS

- Winner, Buchalter Cosmology Prize (2nd Prize) | UCL press release *2023*
  International award recognising *'ground-breaking theoretical, observational, or experimental work in cosmology that has the potential to produce a breakthrough advance in our understanding'*
- Honorable Mention, GWIC-Braccini Thesis Prize | nominated for three further thesis prizes *2022*
- Best Student Talk Prize at BritGrav 21, sponsored by IoP Publishing *2021*
  Corresponding paper published in *Classical and Quantum Gravity* as an invited submission
- King's Education Award for 'extraordinary contributions' to teaching *2020*
  Also nominated as a 'Rising Star' in 2019 (only PhD student nominee in my department)
- Bateman Scholar of Trinity Hall (Cambridge), recognising 'excellent exam results' *2017*

RESPONSIBILITIES

Leadership roles in scientific collaborations and networks

- Co-lead organiser, *CamGW* — the Cambridge gravitational-wave network *2025–present*
  Initiated an inter-departmental network to connect gravitational-wave researchers across Cambridge; organised a successful programme of meetings with external speakers and 40+ attendees
- Co-organiser, *GW:UK* *2025–present*
  Cambridge representative for the UK's national network of gravitational-wave researchers
- Group lead, *GUEST (Gravitational Universe Exploration with Satellite Tracking)* *2025–present*
  Leading a working group to develop the science case for a proposed F-class ESA mission

Organisation of conferences and seminars

- Co-organiser, *Kavli Astrophysics Symposium* — LOC member for international conference *2026*
- Lead organiser, Kavli Science Focus Meeting on *"Cosmology in the Lab"* *2026*
  Organised a two-day interdisciplinary workshop bringing together theorists and experimentalists working in the area of quantum technologies for fundamental physics
- Lead organiser, *UCL Cosmology/Extragalactic Seminars* *2022–2023*
  Developed a programme of in-person talks from what had previously been an online-only event due to COVID-19; secured departmental funding for speaker expenses and group lunches
- Co-organiser, *London Cosmology Discussion Meetings (LCDM)* *2018–2021*
  Coordinated between five institutions to organise meetings at the Royal Astronomical Society
- Lead organiser, *Theoretical Particle Physics and Cosmology (TPPC) Journal Club* *2021*
- Lead organiser, *TPPC Gravity Meetings* *2018–2020*
  Initiated a regular series of meetings with internal and external speakers on gravitational physics

Refereeing and review work

- Referee for seven research grants, including two ERC Consolidator Grants, *2019–present*
  one NSF Research Grant, one EPSRC Research Grant, one EPSRC Stephen Hawking Fellowship, one Royal Society Dorothy Hodgkin Fellowship, and one grant for the Polish National Science Centre
- Referee for 32 articles in Physical Review Letters, Nature Astronomy, *2020–present*
  Astrophysical Journal Letters, Physical Review D, Journal of Cosmology and Astroparticle Physics, European Physical Journal C, The Astronomical Journal, and Universe

Committees and institutional service

- Elector, Selwyn College Mastership Election *2024–2025*
  Contributed to interviews and selection process for the new head of my Cambridge college
- Selwyn College Governing Body and IT and Data Committee *2024–present*
- Student representative, KCL Physics Department Research Committee *2019–2021*

TEACHING AND SUPERVISION

University of Cambridge *2024–present*

- Delivered three 1hr lectures for 3rd-year *Relativity* (Part II Physics/Astrophysics); student feedback highlighted my 'very clear and engaging lecturing'
- Research project supervisor of two MSci students: Teymour Taj and Felix Bowden
- Delivered small-group 'supervision' teaching in *Relativity*, *Statistical Physics*, and *Principles of Quantum Mechanics* for 3rd-year Astrophysicists

University College London *2021–2024*

- Lead supervisor of two MSc students: Phoebe Routh (distinction, now a PhD student in Glasgow) and David Moody (distinction and awarded departmental prize, now working in data science)
- Teaching assistant for 3rd-year *Physical Cosmology*: developed problem sets, delivered problem-solving tutorials, and contributed to marking of final exams

King's College London *2017–2021*

- Winner of a 2020 King's Education Award, recognising 'extraordinary contributions' to teaching
- 'Rising Star' nominee in the 2019 King's Education Awards (only PhD student nominee in Physics)
- Co-wrote lecture notes for 3rd-year *General Relativity and Cosmology*
- Examples class demonstrator for numerous courses, including 4th-year *Astroparticle Cosmology*, 3rd-year *General Relativity and Cosmology*, 2nd-year *Astrophysics*, 1st-year *Mathematics for Physicists*, ...

SELECTED PUBLIC ENGAGEMENT

- Interviewed on *'Unexpected Elements'* from the BBC World Service *2025*
- Contributed to the art/science exhibition *'Cosmic Titans'* at Lakeside Arts, Nottingham *2025*
  Interacted with artist Conrad Shawcross RA to provide input for his piece *'The Blind Proliferation'*; gave an interview exploring the science behind Conrad's work, featured in the exhibition and online
- My research and simulations featured in the documentary *'Do we live in a multiverse?'* *2024*
  Aired on French and German TV and more than 5 million combined views on YouTube
- Led maths outreach activities at a local school as part of the Cambridge NRICH project *2024*
- Invited speaker for the Cambridge Astronomical Association *2024*
- YouTube video interview on *'Early Universe Cosmology in the Lab'* *2023*
- Outreach talk for alumni of UCL's 'Introduction to Astronomy' course *2023*
- Participated in five interviews for media pieces on my paper *2022*
  'Bridging the μHz gap in the gravitational-wave landscape with binary resonance'
- Maths and physics tutor at Open Tutors London *2017–2020*
  Co-initiated tutoring programme for University of London students from under-represented groups
- Ran an interactive exhibit on Dark Matter at Science Gallery London *2019*

SOFTWARE AND NUMERICS

- Author of Fortran lattice field theory code `lattice-fvd` and Python code `gw-resonance`
- Experience with advanced numerical methods including, e.g., Fourier and Chebyshev pseudospectral methods and symplectic integration
- Extensive experience in Unix environments (Ubuntu/MacOS), including in HPC settings; extensive Python experience (object-oriented programming; data handling and visualisation; `JAX`, `Jupyter`, `NumPy`, `SciPy`, `h5py`, `Astropy`, `healpy`, `sympy`, ...); other languages and software include Fortran, C++, Mathematica, Git, MATLAB, SageMath, SQL, LaTeX (including TikZ), ...

SELECTED PUBLICATIONS

Citation statistics as of 11 March 2026 (data from INSPIRE-HEP):

Lead-author only	17 papers, 772 citations, h-index = 13
Short author list only	30 papers, 1,545 citations, h-index = 18
All publications	112 papers, 41,677 citations, h-index = 71

Lead-author papers (author list is ordered alphabetically in some cases)

L1. **ACJ**, H. V. Peiris, and A. Pontzen, *Bubbles in a box: Eliminating edge nucleation in cold-atom simulators of vacuum decay* (2025), Phys. Rev. A **112**, 023318, arXiv:2504.02829 [cond-mat.quant-gas]

L2. **ACJ**, I. G. Moss, T. P. Billam, Z. Hadzibabic, H. V. Peiris, and A. Pontzen, *Generalized cold-atom analogues for vacuum decay* (2024), Phys. Rev. A **110**, L031301, arXiv:2311.02156 [cond-mat.quant-gas] | Letter

L3. **ACJ**, J. Braden, H. V. Peiris, A. Pontzen, M. C. Johnson, and S. Weinfurtner, *Analog vacuum decay from vacuum initial conditions* (2024), Phys. Rev. D **109**, 023506, arXiv:2307.02549 [cond-mat.quant-gas] | Editor's Suggestion

L4. A. K.-W. Chung, **ACJ**, J. D. Romano, and M. Sakellariadou, *Targeted search for the kinematic dipole of the gravitational-wave background* (2022), Phys. Rev. D **106**, 082005, arXiv:2208.01330 [gr-qc]

L5. M. R. Mosbech, **ACJ**, S. Bose, C. Boehm, M. Sakellariadou, and Y. Y. Y. Wong, *Gravitational-wave event rates as a new probe for dark matter microphysics* (2023), Phys. Rev. D **108**, 043512, arXiv:2207.14126 [astro-ph.CO]
Co-lead author with Markus Mosbech; I developed the core idea and led $\sim 50\%$ of the analysis
Featured in a Royal Astronomical Society press release at the 2023 National Astronomy Meeting

L6. D. Blas and **ACJ**, *Bridging the μHz gap in the gravitational-wave landscape with binary resonance* (2022), Phys. Rev. Lett. **128**, 101103, arXiv:2107.04601 [astro-ph.CO]
Awarded a Buchalter Cosmology Prize (2nd Prize), recognising 'potential for remarkable impact'
Altmetric attention score of 395, in the top 0.3% of all publications ever tracked by Altmetric
Featured in the *Daily Express*, *Physics* magazine, *Big Think*, *SYFY wire*, and 40+ other outlets

L7. D. Blas and **ACJ**, *Detecting stochastic gravitational waves with binary resonance* (2022), Phys. Rev. D **105**, 064021, arXiv:2107.04063 [gr-qc]

L8. **ACJ** and M. Sakellariadou, *Nonlinear gravitational-wave memory from cusps and kinks on cosmic strings* (2021), Class. Quant. Grav. **38**, 165004, arXiv:2102.12487 [gr-qc]
Invited submission to CQG as winner of the Best Student Talk Prize at BritGrav 21

L9. **ACJ** and M. Sakellariadou, *Primordial black holes from cusp collapse on cosmic strings* (2020), arXiv:2006.16249 [astro-ph.CO]

L10. **ACJ**, J. D. Romano, and M. Sakellariadou, *Estimating the angular power spectrum of the gravitational-wave background in the presence of shot noise* (2019), Phys. Rev. D **100**, 083501, arXiv:1907.06642 [astro-ph.CO]

L11. **ACJ** and M. Sakellariadou, *Shot noise in the astrophysical gravitational-wave background* (2019), Phys. Rev. D **100**, 063508, arXiv:1902.07719 [astro-ph.CO]

L12. **ACJ**, R. O'Shaughnessy, M. Sakellariadou, and D. Wysocki, *Anisotropies in the astrophysical gravitational-wave background: The impact of black hole distributions* (2019), Phys. Rev. Lett. **122**, 111101, arXiv:1810.13435 [astro-ph.CO]

L13. **ACJ**, A. G. A. Pithis, and M. Sakellariadou, *Can we detect quantum gravity with compact binary inspirals?* (2018), Phys. Rev. D **98**, 104032, arXiv:1809.06275 [gr-qc]

L14. **ACJ**, M. Sakellariadou, T. Regimbau, and E. Slezak, *Anisotropies in the astrophysical gravitational-wave background: Predictions for the detection of compact binaries by LIGO and Virgo* (2018), Phys. Rev. D **98**, 063501, arXiv:1806.01718 [astro-ph.CO]

L15. **ACJ** and M. Sakellariadou, *Anisotropies in the stochastic gravitational-wave background: Formalism and the cosmic string case* (2018), Phys. Rev. D **98**, 063509, arXiv:1802.06046 [astro-ph.CO]
Featured in PRD's 'kaleidoscope' for Sep 2018

Other selected papers (with summary of my main contributions)

O1. Y. Zhang, F. Wang, P. Wong, **ACJ**, K. Konstantinou, J. Thywissen, C. Eigen, and Z. Hadzibabic, *Quantum Simulation of Massive Relativistic Fields in 2+1 Dimensions* (2026), arXiv:2603.08840 [cond-mat.quant-gas] | First experimental realisation of relativistic 2+1D Sine-Gordon model, enabled by my earlier work demonstrating its viability [L2]; significant contributions to theory, conceptualisation, and experimental design throughout the project

O2. J. W. Foster, D. Blas, A. Bourgoin, A. Hees, M. Herrero-Valea, **ACJ**, and X. Xue, *Prospects for gravitational wave and ultra-light dark matter detection with binary resonances beyond the secular approximation* (2025), arXiv:2504.16988 [gr-qc]
Significant conceptual and methodological contributions throughout project

O3. J. W. Foster, D. Blas, A. Bourgoin, A. Hees, M. Herrero-Valea, **ACJ**, and X. Xue, *Discovering μHz gravitational waves and ultra-light dark matter with binary resonances* (2025), arXiv:2504.15334 [astro-ph.CO] | Significant conceptual and methodological contributions throughout project

O4. A. Abac *et al.*, *The Science of the Einstein Telescope* (2025), arXiv:2503.12263 [gr-qc]
Coordinated and led writing of section on cosmic strings for large collaboration white paper

O5. L. Zwick, D. Soyuer, D. J. D'Orazio, D. O'Neill, A. Derdzinski, P. Saha, D. Blas, **ACJ**, L. Z. Kelley, *Bridging the micro-Hz gravitational wave gap via Doppler tracking with the Uranus Orbiter and Probe Mission: Massive black hole binaries, early universe signals and ultra-light dark matter* (2024), arXiv:2406.02306 [astro-ph.HE] | Led analysis of stochastic gravitational-wave signals and implications for early-Universe physics, wrote corresponding sections of paper

O6. N. Kouvatsos, **ACJ**, A. I. Renzini, J. D. Romano, M. Sakellariadou, *Unbiased estimation of gravitational-wave anisotropies from noisy data* (2023), arXiv:2312.09110 [astro-ph.CO]
Proposed new analysis method that has since been adopted by the LVK, and led theoretical work; informal supervision of PhD student (N. Kouvatsos)

O7. M. Branchesi *et al.*, *Science with the Einstein Telescope: a comparison of different designs* (2023), JCAP **07**, 068, arXiv:2303.15923 [gr-qc] | Contributed to stochastic background sensitivity analysis for different Einstein Telescope configurations, guiding further development of the proposal

O8. S. Gasparrotto, R. Vicente, D. Blas, **ACJ**, and E. Barausse, *Can gravitational-wave memory help constrain binary black-hole parameters? A LISA case study* (2023), Phys. Rev. D **107**, 124033, arXiv:2301.13228 [gr-qc] | Significant conceptual and methodological contributions; informal supervision of PhD student (S. Gasparrotto)

O9. P. Auclair *et al.* (LISA Cosmology Working Group), *Cosmology with the Laser Interferometer Space Antenna* (2022), Living Rev. Rel. **26**, 5, arXiv:2204.05434 [astro-ph.CO] | LISA white paper; contributed to section on cosmic strings, led analysis of related gravitational-wave anisotropies

O10. A. I. Renzini, B. Goncharov, **ACJ**, and P. M. Meyers, *Stochastic Gravitational-Wave Backgrounds: Current Detection Efforts and Future Prospects* (2022), Galaxies **10**, 34, arXiv:2202.00178 [gr-qc] Invited review article; major contributions to sections on gravitational-wave theory and sources

O11. N. Bartolo *et al.* (LISA Cosmology Working Group), *Probing Anisotropies of the Stochastic Gravitational Wave Background with LISA* (2022), JCAP **11**, 009, arXiv:2201.08782 [astro-ph.CO] LISA review paper; coordinator for 'topological defects' section, with further contributions to 'astrophysical sources' section

O12. P. Auclair, J. J. Blanco-Pillado, D. G. Figueroa, **ACJ**, M. Lewicki, M. Sakellariadou, S. Sanidas, L. Sousa, D. A. Steer, J. M. Wachter, and S. Kuroyanagi (LISA Cosmology Working Group), *Probing the gravitational wave background from cosmic strings with LISA* (2019), JCAP **04**, 034, arXiv:1909.00819 [astro-ph.CO] | LISA review paper; significant writing contributions throughout

O13. B. P. Abbott *et al.* (LIGO, Virgo), *Directional limits on persistent gravitational waves using data from Advanced LIGO's first two observing runs* (2019), Phys. Rev. D **100**, 062001, arXiv:1903.08844 [gr-qc] | Led interpretation of observational results in the context of cosmological and astrophysical source models, wrote corresponding section

O14. B. P. Abbott *et al.* (LIGO, Virgo), *Search for the isotropic stochastic background using data from Advanced LIGO's second observing run* (2019), Phys. Rev. D **100**, 061101, arXiv:1903.02886 [gr-qc] Rapid communication | Led and wrote section on implications for cosmic string models

INVITED TALKS

Total of 50 invited talks across three continents

- **Cosmology Seminar** — *Apr 2026*
  Queen Mary University of London
- **Particle Theory and Cosmology Seminar** — *Mar 2026*
  Swansea University
- **Brout-Englert-Lemaître Seminar** — *Feb 2026*
  Brussels
- **Quantum Light and Matter Seminar** — *Nov 2025*
  Durham University
- **Particle Cosmology Seminar** — *Oct 2025*
  Imperial College London
- **Numerical Simulations of Early Universe Sources of Gravitational Waves** — *Aug 2025*
  Nordita, Stockholm
- **Cosmology Seminar** — *Jul 2025*
  CERN, Geneva
- **Gravitational Wave Probes of Physics Beyond Standard Model** — *Jun 2025*
  University of Warsaw
- **Multiverse from Home** — *Jun 2025*
  Online

- **WE-Heraeus Seminar: New Windows on the Universe** *May 2025*
  Kitzbühel, Austria
- **QSimFP Community Workshop** *Mar 2025*
  University of Nottingham
- **Early Universe from Home** *Feb 2025*
  Online
- **Theoretical Particle Physics Seminar** *Feb 2025*
  University of Sussex
- **Joint seminar, LMU Cosmology and MPI Quantum Optics** *Jan 2025*
  Ludwig Maximilian University of Munich
- **General Relativity Seminar** *Dec 2024*
  DAMTP, University of Cambridge
- **Cosmology Journal Club** *Nov 2024*
  University of Padova (online)
- **Particle Cosmology Seminar** *Nov 2024*
  University of Nottingham
- **Cosmology/Extragalactic Seminar** *Oct 2024*
  University College London
- **Cambridge-LMU Meeting** *Oct 2024*
  Kavli Institute for Cosmology, University of Cambridge
- **GEMMA2 Workshop** *Sep 2024*
  Sapienza University of Rome
- **Majorana-Raychaudhuri Seminar** *Sep 2024*
  Kolkata/Salerno (online)
- **Cold atoms and molecules for fundamental physics** *Jul 2024*
  Cambridge
- **4th EuCAPT Symposium** *May 2024*
  CERN, Geneva
- **British Applied Mathematics Colloquium (BAMC)** *Apr 2024*
  Newcastle University
- **Cosmology Lunch Seminar** *Feb 2024*
  DAMTP, University of Cambridge
- **Gravitational-Wave Group Meeting** *Jan 2024*
  Institute of Cosmology and Gravitation (ICG), University of Portsmouth
- **Oberthaler Group Seminar** *Nov 2023*
  Kirchhoff Institute for Physics (KIP), Heidelberg
- **Cosmology from Home** *Jul 2023*
  Online
- **Quantum Simulators for Fundamental Physics Workshop** *Jun 2023*
  Perimeter Institute for Theoretical Physics, Waterloo, Canada
- **Astrophysics Seminar** *May 2023*
  University of Leicester

- **Cosmology Seminar** *May 2023*
  Beecroft Institute, Oxford
- **Theory Group Seminar** *May 2023*
  Astroparticle and Cosmology Laboratory (APC), Paris
- **Cosmology and Relativity Seminar** *Apr 2023*
  Queen Mary University of London
- **London Gravity Meeting** *Mar 2023*
  Royal Society, London
- **'Dark Matters' Workshop** *Nov 2022*
  Université Libre de Bruxelles (ULB)
- **London-Oldenburg Relativity Seminar** *Nov 2022*
  University College London/University of Oldenburg (online)
- **ICTP-AP Seminar** *Sep 2022*
  International Centre for Theoretical Physics, Asia-Pacific (online)
- **Quantum Simulators for Fundamental Physics Workshop** *Sep 2022*
  Science Gallery London
- **'Gravitational-Wave Orchestra' Workshop** *Sep 2022*
  Université Catholique de Louvain, Belgium
- **Theory Group Seminar** *May 2022*
  Institute of High-Energy Physics (IFAE), Barcelona
- **Quantum Technology Seminar** *May 2022*
  London Centre for Nanotechnology, University College London
- **Cosmology/Extragalactic Seminar** *Nov 2021*
  University College London
- **Theory Group Seminar** *Oct 2021*
  Astroparticle and Cosmology Laboratory (APC), Paris
- **Ibarra Group Seminar** *Jul 2021*
  Technical University of Munich (online)
- **Gravitational Wave Probes of Physics Beyond the Standard Model** *Jul 2021*
  University of Warsaw (online)
- **London Cosmology Discussion Meeting (LCDM)** *Dec 2020*
  Royal Astronomical Society, London (online)
- **Theoretical Cosmology Seminar** *May 2020*
  Institute of Cosmology and Gravitation (ICG), University of Portsmouth (online)
- **Cosmology Seminar** *May 2020*
  Beecroft Institute, University of Oxford (online)
- **London Cosmology Discussion Meeting (LCDM)** *Feb 2020*
  Royal Astronomical Society, London
- **Gravitational Wave Probes of Fundamental Physics** *Nov 2019*
  EuCAPT workshop, Amsterdam
- **Seminar** *Feb 2019*
  Virtual Institute of Astroparticle Physics (online)